

Multiple Choice

1. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: B - False, False

2. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: C - True, False

3. **Answer:** D

Options: A) $(S, +, \times)$ is closed under addition modulo 10.; B) $(S, +, \times)$ is closed under multiplication modulo 10.; C) $(S, +, \times)$ has an identity under addition modulo 10.; D) $(S, +, \times)$ has no identity under multiplication modulo 10.

Note: Correct option: D - $(S, +, \times)$ has no identity under multiplication modulo 10.

4. **Answer:** C

Options: A) 27; B) 28; C) 35; D) 37

Note: Correct option: C - 35

5. **Answer:** C

Options: A) 4; B) 6; C) 12; D) 24

Note: Correct option: C - 12

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'transitive'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'If A is false, then B is false.'

Long Form Response

11. **Answer:** Let

$$t = \frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}.$$

Then $x = 3t + 2$, $y = 4t - 1$, and $z = 12t + 2$. Substituting into $x - y + z = 5$ gives us

$$(3t + 2) - (4t - 1) + (12t + 2) = 5.$$

Solving, we find $t = 0$. Thus, $(x, y, z) = (2, -1, 2)$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** First, we rearrange the equation so that one side is a quadratic written in the usual way and the other side is 0. We can do this by subtracting $20x$ from each side (and reordering the terms): $5x^2 - 20x + 18 = 0$ *This doesn't factor in any obvious way, so we apply the quadratic formula, which,*
- $$\frac{-(-20) \pm \sqrt{(-20)^2 - 4(5)(18)}}{2(5)} = \frac{20 \pm \sqrt{400 - 360}}{10}$$
- $$= \frac{20 \pm \sqrt{40}}{10}$$
- $$= 2 \pm \frac{\sqrt{40}}{10}. \text{ We observe that } \sqrt{40} \text{ is between 6 and 7, so } \frac{\sqrt{40}}{10} \text{ is between 0.6 and 0.7. Thus one}$$
- of our solutions is between 1.3 and 1.4, while the other is between 2.6 and 2.7. Rounding each solution to the nearest integer, we get 1 and 3, the product of which is 3\$.

Note: Students should show all steps in their mathematical solution.

End of Answer Key